

Exercise 2: Data Preprocessing

Submission: Sunday, 29th March, 23:59h

Task 1: Data Preprocessing

1 points

In this task, you will perform data cleaning and preprocessing on the provided dataset. You may **reuse** and extend the application you created in Exercise 1.

- a. **Download** the dataset from OLAT. The dataset contains the following variables:
 - **Area:** area name
 - **Year:** year
 - **Item:** crop type
 - **Yield_hg_ha:** crop yield (hectograms per hectare)
 - **Rainfall_mm:** average annual rainfall (millimeters)
 - **Temp:** average annual temperature (°C)
 - **Pesticides_ton_ha:** total pesticide usage (ton per hectare)
 - **Continent:** continent of the area
- b. Detect all missing values in the provided dataset. Report the number of missing values for each column. Apply appropriate strategies to handle missing values. Clearly explain and justify your chosen imputation strategy. Submit your answers in a **.txt** or **.pdf** file.

Task 2: Data analysis

In this task, we will add the visualization of the statistics (**Mean, Median, Variance**) of the different variables of the dataset.

- a. Create a line chart to show how a selected statistical measure varying over time:
 - The x-axis must represent **Year**.
 - The y-axis must represent the selected measure of the selected attribute.
 - The statistic must be computed using all areas within the selected continent.

2 point

For example, if the user selects **Europe, Rainfall_mm, mean**, the plot should show the mean rainfall across all European areas for each year.

- b. Create a set of box plots that display the distribution of the selected attribute:
- The x-axis must represent the different **Areas**.
 - The box plots should show the distribution of the selected attribute across countries within the selected continent.

For example, if **Europe** and **Rainfall_mm** are selected, each box plot should show the rainfall distribution (across all years) for one European area.

- c. For **Yield_hg_ha**, the visualization must additionally reflect differences between crops:
- The statistical analysis should distinguish between different **Items (crops)**.
 - The visualization should clearly show the distribution of yields across different crops.

For all other attributes (**Rainfall_mm**, **Temp**, **Pesticides_ton_ha**), aggregation should reflect one value per country per year, and no crop-level differentiation is required.

- d. Implement the dropdown menus for users to select Continent, Attribute, and Statistics (**Mean, Median, Variance**). The two plots above should change dynamically according to the user's selection.

Hand-in

Please hand in your solution in OLAT as a **.zip** file until **Sunday, 29th March, 23:59h**. The **.zip** file should include the following:

- Your **python code** with the runnable final application in a single file with a reasonable amount of comments explaining your implementation

Note: You are **not** allowed to use **other libraries** despite the ones stated in environment.yml.

- Include at least one **screenshot** demonstrating your running application.

Note: Submissions without screenshots will result in a **0.5 point deduction**.

- Required **.txt** or **.pdf** file. In case there are additional remarks, include them as a **.txt** or **.pdf** file too.
- Please hand-in you own unique solutions. We do not accept copied solutions or solutions created by or with ChatGPT!